

Sample for Quiz 1

DATA 412/612

1. Create the following matrix:

```
A <- 10*t(matrix(99:0, 10,10))  
A
```

	[,1]	[,2]	[,3]	[,4]	[,5]	[,6]	[,7]	[,8]	[,9]	[,10]
[1,]	990	980	970	960	950	940	930	920	910	900
[2,]	890	880	870	860	850	840	830	820	810	800
[3,]	790	780	770	760	750	740	730	720	710	700
[4,]	690	680	670	660	650	640	630	620	610	600
[5,]	590	580	570	560	550	540	530	520	510	500
[6,]	490	480	470	460	450	440	430	420	410	400
[7,]	390	380	370	360	350	340	330	320	310	300
[8,]	290	280	270	260	250	240	230	220	210	200
[9,]	190	180	170	160	150	140	130	120	110	100
[10,]	90	80	70	60	50	40	30	20	10	0

2. Find the sum of the entries of the even columns of the above matrix.

```
sum(A[,2],A[,4],A[,6],A[,8],A[,10])  
# [1] 24500
```

3. Suppose you have the following data frame:

```
df <- data.frame(  
  Name = c("Johnny", "Alisson", "Andy", "Sarah", "Alex"),  
  Age = c(25, 30, 28, 22, 32),  
  Annual_Income = c(50000, 60000, 55000, 48000, 70000),  
  Account_Balance = c(1700, 2000, 1800, 900, 2500.90)  
)  
df
```

	Name	Age	Annual_Income	Account_Balance
1	Johnny	25	50000	1700.0
2	Alisson	30	60000	2000.0
3	Andy	28	55000	1800.0
4	Sarah	22	48000	900.0
5	Alex	32	70000	2500.9

- What is the age average of people who get more than \$51k in a year.

```
mean(df$Age[df$Annual_Income > 51000])  
# [1] 30
```

- What are their names?

```
df$Name[df$Annual_Income > 51000]  
# [1] "Alisson" "Andy" "Alex"
```

4. Given a data frame `df` with only positive values, write a function that check if the number of columns in it is less than equal to 1 or it has negative values at all. If yes, it returns `-1`. If not, the function compares the geometric means (using the pipe operator) of the first and the second columns and produces 1 if that is correct, otherwise 0. Try your function on these inputs `data.frame(a=c(9,1,8))`, `data.frame(a=c(9,9,8), b=c(9,7,6))`, and `data.frame(a=c(9,1,8), b=c(9,17,6))`.

```
library(tidyverse)

my_df_function <- function(df) {
  if (length(df) <= 1 | any(df < 0)) {
    return(-1)
  } else {
    df[,1] %>%
      log() %>%
      mean() %>%
      exp() -> geo_mean_1

    df[,2] %>%
      log() %>%
      mean() %>%
      exp() -> geo_mean_2

    if (geo_mean_1 >= geo_mean_2) {
      return(1)
    } else {
      return(0)
    }
  }
}

df <- data.frame(a=c(-9,1,8), b=c(-4,3,18))
my_df_function(df) # [1] -1

df1 <- data.frame(a=c(9,9,8), b=c(9,7,6))
my_df_function(df1) # [1] 1

df2 <- data.frame(a=c(9,1,8), b=c(9,17,6))
my_df_function(df2) # [1] 0
```